

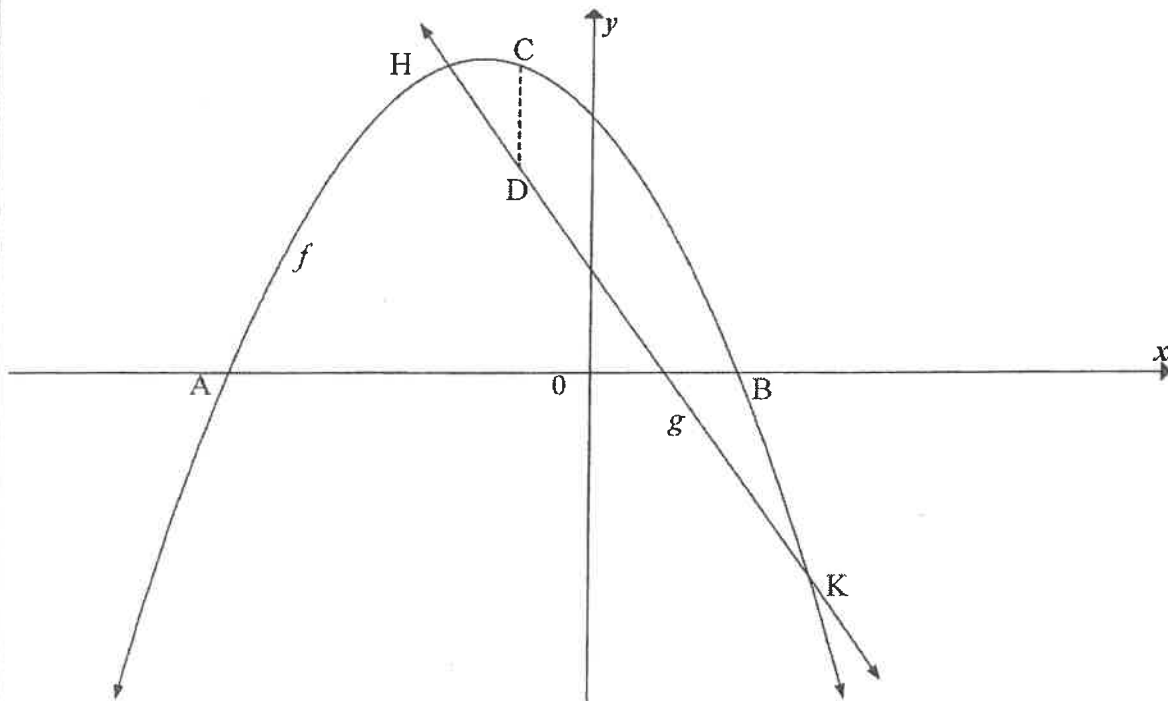
6

Functions Paper 1

PARABOLAS AND LINES

QUESTION 1

The sketch below shows the graphs of $g(x) = -12x + 12$ and $f(x) = -3x^2 - 9x + 30$. A and B are the x -intercepts of f and C is a point on f . D is a point on g such that CD is parallel to the y -axis. H and K are the points of intersection of f and g .

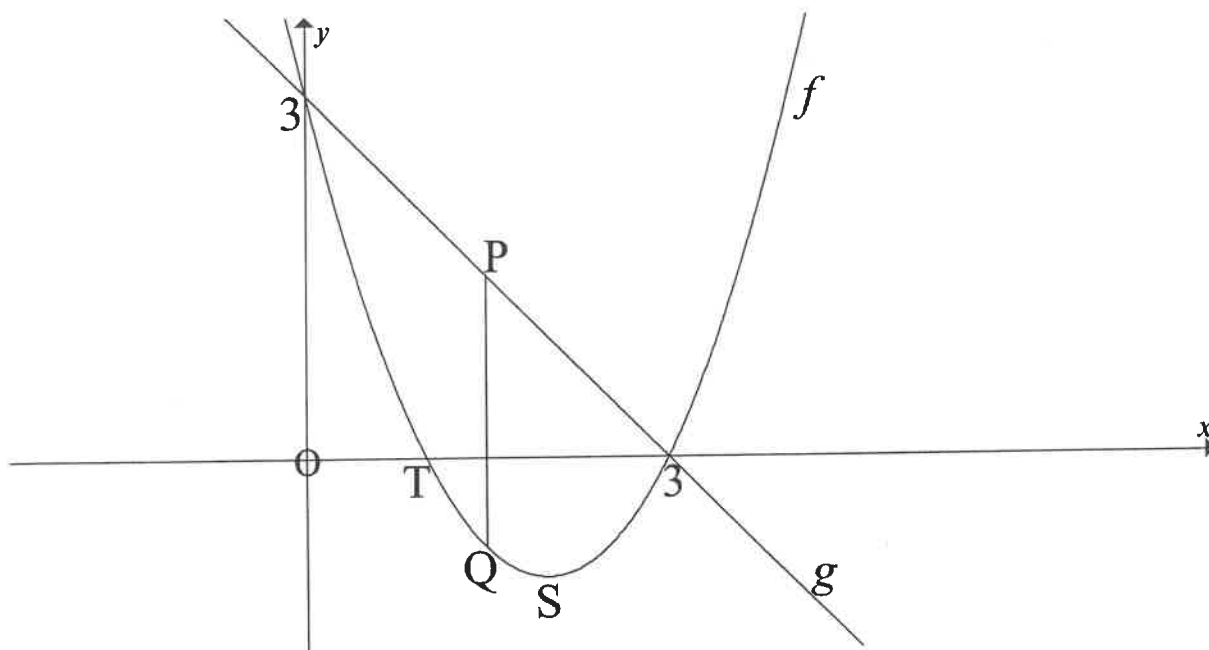


- 1.1. Determine the length of AB. (4)
- 1.2. Determine the coordinates of K. (5)
- 1.3. Determine the values of x for which $f(x) - g(x) \leq 0$. (3)
- 1.4. Determine the maximum length of CD for $-2 \leq x \leq 3$. (5)

[17]

QUESTION 2

The sketch shows the graphs of $f(x) = x^2 - 4x + 3$ and $g(x) = 3 - x$:



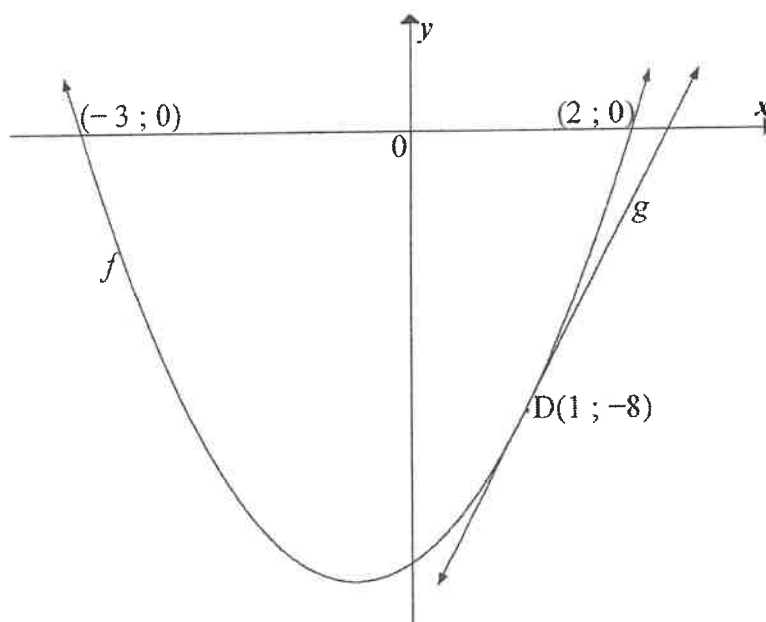
- 2.1. Determine the co-ordinates of T. (4)
- 2.2. Write down the co-ordinates of S, the turning point of f . (3)
- 2.3. PQ is variable line segment connecting a point on f to a point on g for $x \in [0; 3]$. $PQ \perp x$ -axis. Determine the co-ordinates of P for which PQ has a maximum length. *Show your workings and reasoning.* (5)
- 2.4. If the graph of $y = f(x + p) + q$ has its turning point at the origin, determine the values of p and q . (2)
- 2.5. For which values of x is $f(x) < g(x)$? (2)

QUESTION 3

The graphs of $f(x) = ax^2 + bx + c$; $a \neq 0$ and $g(x) = mx + k$ are drawn below.

$D(1 ; -8)$ is a common point on f and g .

- f intersects the x -axis at $(-3 ; 0)$ and $(2 ; 0)$.
- g is the tangent to f at D .



- 3.1. For which value(s) of x is $f(x) \leq 0$? (2)
- 3.2. Determine the values of a , b and c . (5)
- 3.3. Determine the coordinates of the turning point of f . (3)
- 3.4. Write down the equation of the axis of symmetry of h if $h(x) = f(x - 7) + 2$. (2)
- 3.5. Calculate the gradient of g . (3)

Hyperbolas

QUESTION 4

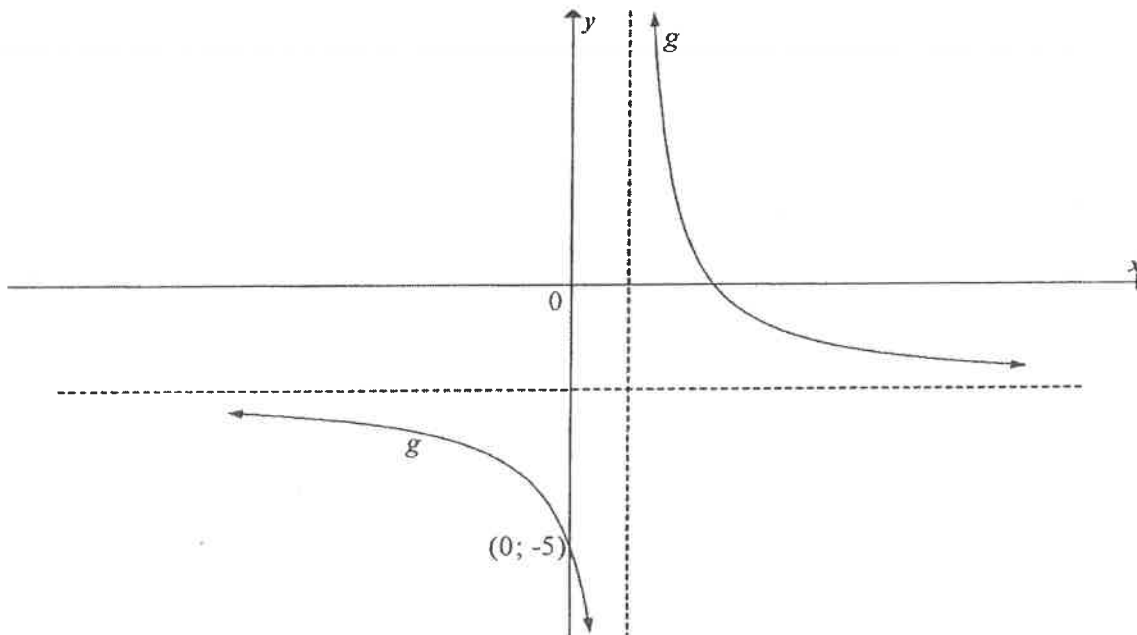
Given: $g(x) = \frac{6}{x+2} - 1$

- 4.1 Write down the equations of the asymptotes of g . (2)
- 4.2 Calculate: (1)
- 4.2.1 The y -intercept of g (2)
- 4.2.2 The x -intercept of g (3)
- 4.3 Draw the graph of g , showing clearly the asymptotes and the intercepts with the axes. (3)
- 4.4 Determine the equation of the line of symmetry that has a negative gradient, in the form $y = \dots$ (3)
- 4.5 Determine the value(s) of x for which $\frac{6}{x+2} - 1 \geq -x - 3$. (2)

[13]

QUESTION 5

The diagram below shows the graph of $g(x) = \frac{a}{x-1} - 2$. The point $(0; -5)$ lies on g .

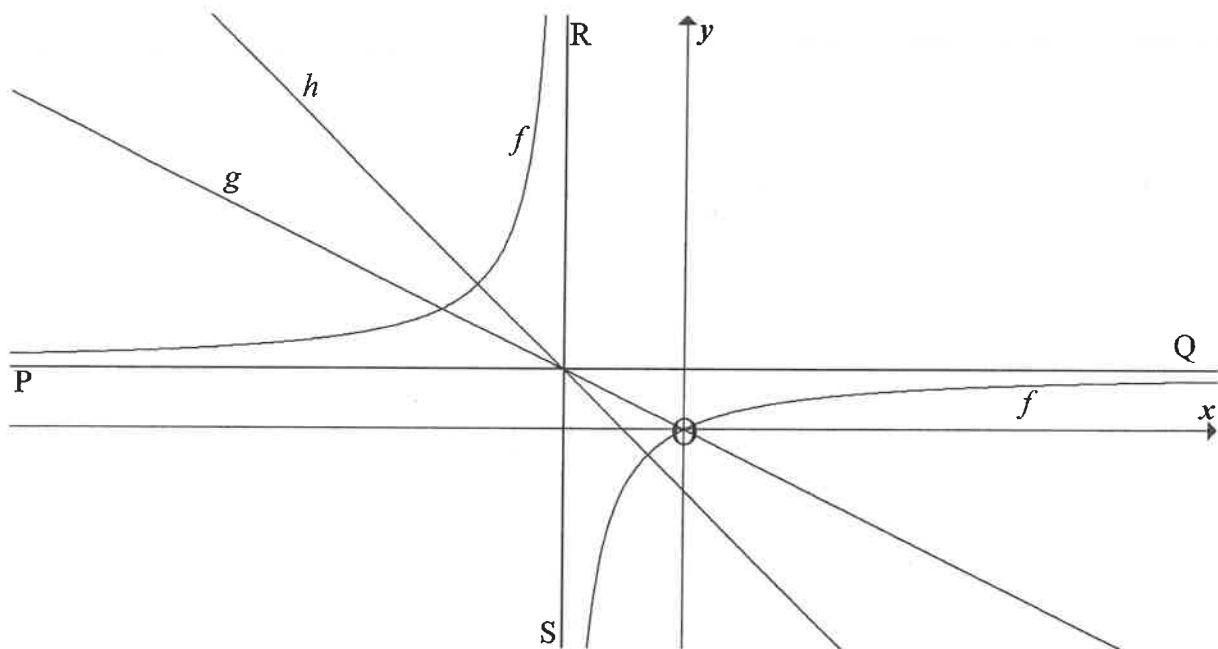


- 5.1. Write down the range of g . (2)
- 5.2. Determine the value of a . (2)
- 5.3. If another function h is defined as $h(x) = g(x-3) + 7$, determine the coordinates of the point of intersection of the asymptotes of h . (3)

[7]

QUESTION 6

The sketch shows the graphs of $f(x) = \frac{k}{x+2} + 1$ and $g(x) = mx$. Lines PQ and RS are asymptotes of f . h is an axis of symmetry of f .



- 6.1 Write down the equation of PQ. (1)
- 6.2 Write down the domain of f . (2)
- 6.3 Determine the equation of h , in the form $h(x) = \dots$ (2)
- 6.4 Given that $s(x) = -f(x) + 3$.
 - 6.4.1 Describe the transformation from the graph of f to the graph of s , in words. (2)
 - 6.4.2 Write down the range of s . (2)
- 6.5 Calculate the values of k and m . (4)

QUESTION 7

Given $f(x) = \frac{x-3}{x+2}$

7.1. What is the domain of f ?

7.2. Determine the co-ordinates of the

7.2.1. x -intercept;

7.2.2. y -intercept
of f .

7.3. Rewrite the equation into the form

$f(x) = \frac{a}{x+p} + q$ and hence write down
the equation of the horizontal asymptote
of f .

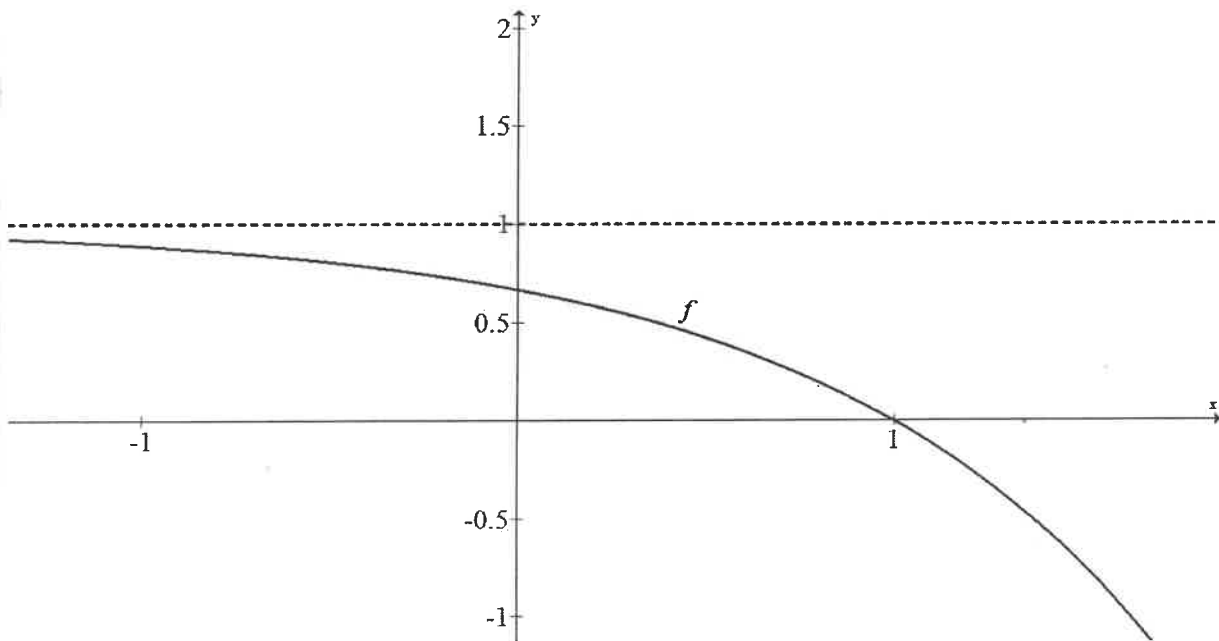
7.4. Sketch the graph of $y = f(x)$.

7.5. Use your graph to solve for x in the
inequality $\frac{x-3}{x+2} \leq 0$.

EXPONENTIAL FUNCTIONS

QUESTION 8

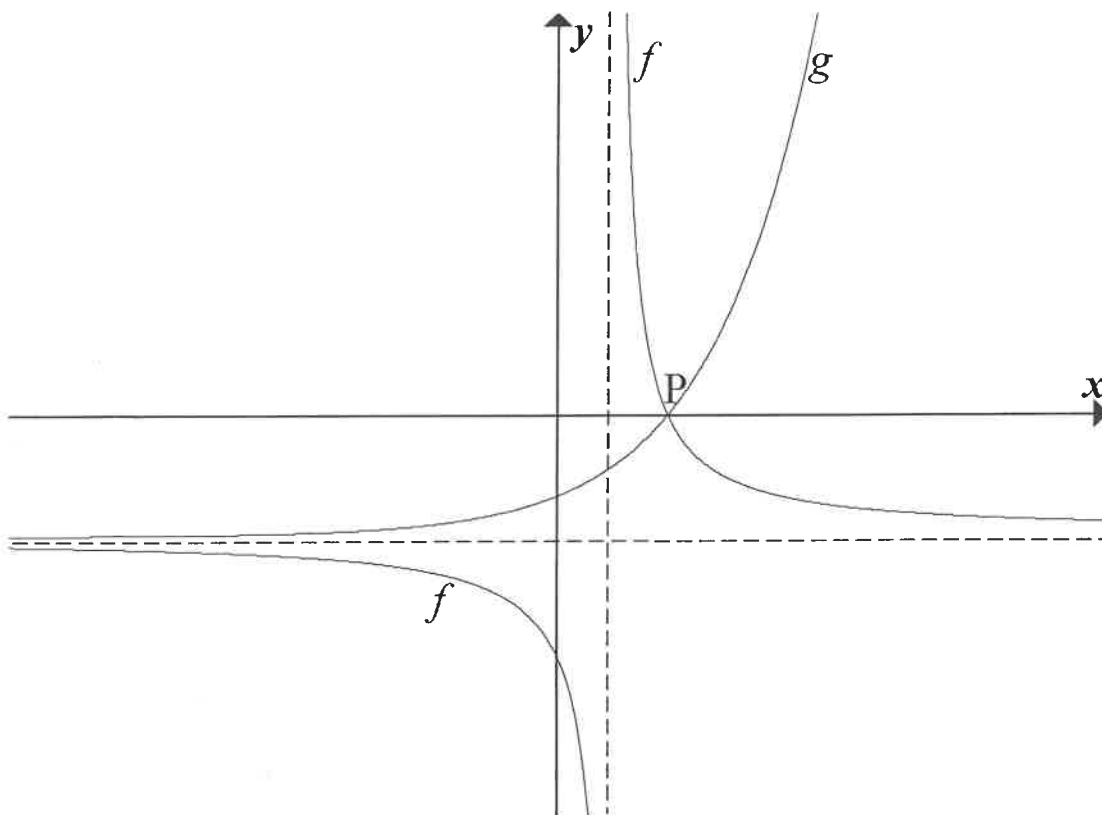
The graph below is of $f(x) = -(3)^{x+p} + q$:



- 8.1 Determine the values of p and q and hence the equation of f . (3)
- 8.2 Write down the equation of k , the translation of f , 3 units down and 2 units to the right. (3)
- 8.3 Write down the equation of l , the reflection of f , about the x -axis. (3)
- [9]

QUESTION 9

The following sketch shows the graphs of $f(x) = \frac{3}{x-1} - 3$ and $g(x) = a^x + b$:



*NB: f and g have the same horizontal asymptote and the same x -intercept.

- 9.1. Write down the equation of the horizontal asymptote of f . (1)
- 9.2. Write down the domain of f . (2)
- 9.3. Determine the values of a and b . (3)
- 9.4. Write down the range of g . (2)
- 9.5. If P is the point (2;0), for which values of x is
 - 9.5.1. $f(x) \geq g(x)$? (2)
 - 9.5.2. $f(x) \cdot g(x) > 0$? (2)

INVERSES AND SYMMETRIES / REFLECTIONS

QUESTION 10

Consider the function: $f(x) = \frac{1}{2}x^2 ; x \leq 0$

- 10.1. Explain the effect that the restriction ($x \leq 0$) has on the function and subsequently on its inverse. (2)
- 10.2. Write down the equation of f^{-1} in the form $f^{-1}(x) = \dots$ (3)
- 10.3. Sketch the graphs of f and f^{-1} on the same set of axes. (6)
- 10.4. If g is the reflection of f in the x -axis, write down the equation of g in the form $g(x) = \dots$ (2)

[13]

QUESTION 11

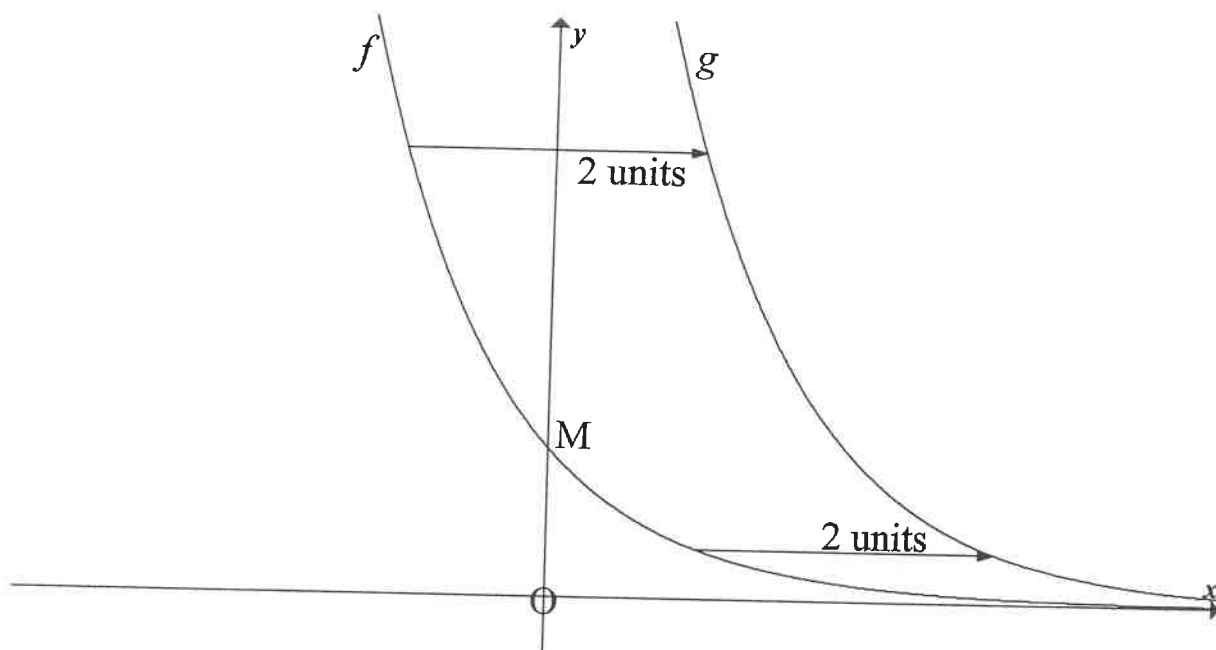
Given: $f(x) = \left(\frac{1}{4}\right)^x$

- 11.1. Determine the value of $f(-2)$. (2)
- 11.2. Write down the equation of $f^{-1}(x)$ in the form $y = \dots$ (2)
- 11.3. Andrew has no idea how to draw the graph of f^{-1} . Explain to Andrew how he may use the graph of f to draw the graph of f^{-1} . (1)
- 11.4. Hence or otherwise, sketch the graph of f^{-1} in your ANSWER BOOK. Clearly indicate ALL intercepts with the axes. (2)
- 11.5. Write down the domain of f^{-1} . (1)
- 11.6. For which values of x will $f^{-1}(x) \geq -2$? (2)
- 11.7. Given: $q = \log_{\frac{1}{4}} \frac{1}{2}$
- 11.7.1. Determine the value of q . (1)
- 11.7.2. Hence or otherwise, determine the coordinates of the point of intersection of f and f^{-1} . (3)

[14]

QUESTION 12

The sketch shows the graphs of $f(x) = a^x$ and $g(x) = 9a^x$:



The graph of g may be obtained by shifting the graph of f two units to the right.

12.1. Calculate the value of a . (3)

12.2. Given that $a = \frac{1}{3}$

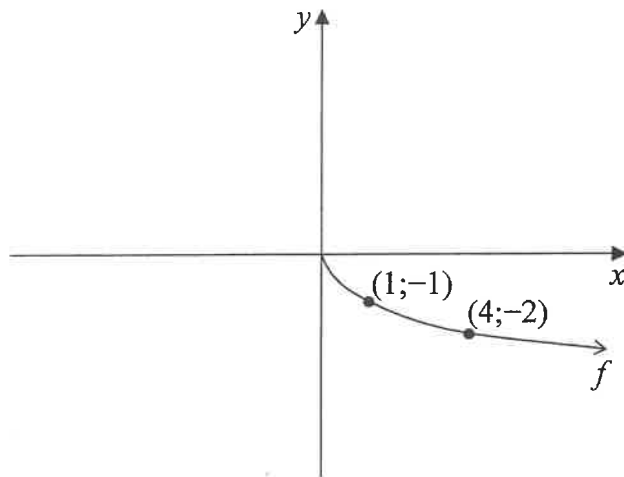
12.2.1. Write down the equation of the inverse of f in the form $f^{-1}(x) = \dots$ (2)

12.2.2. f^{-1} is the mirror image of f in a specific line. Write down the equation of this line. (1)

12.2.3. Write down the value of t if $y = t^x$ is the reflection of $y = f(x)$ in the y -axis. (1)

QUESTION 13

The following drawing shows the graph of $f(x) = -\sqrt{x}$.



g is the reflection of f in the y -axis.

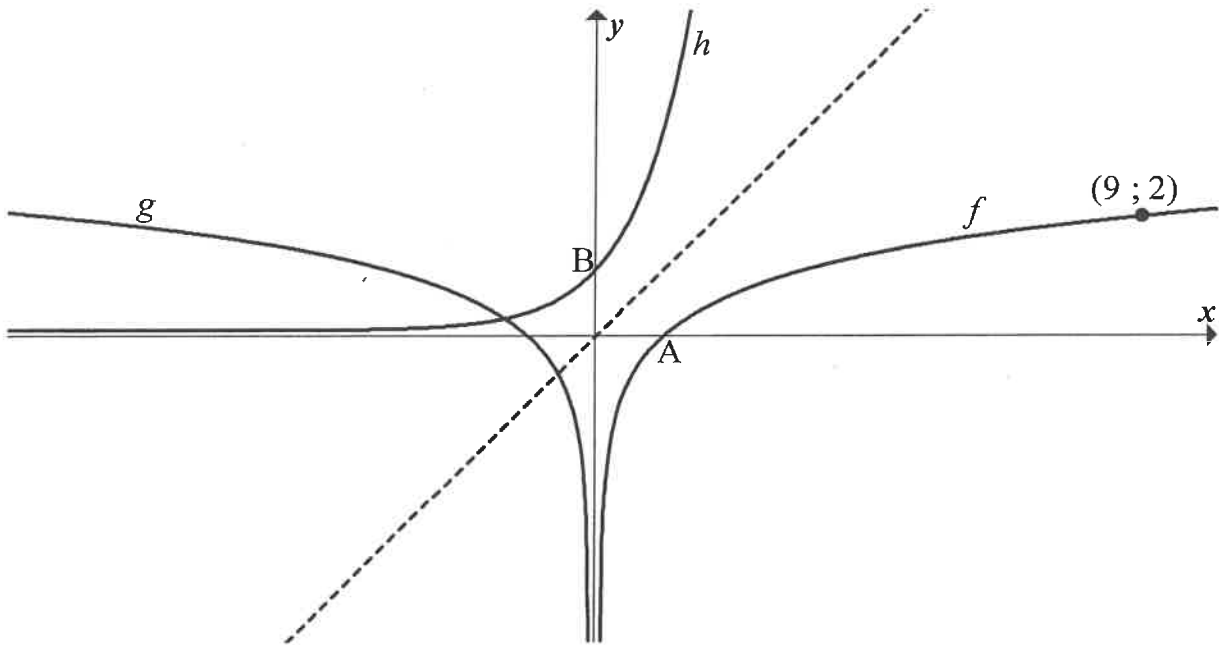
h is the reflection of f in the line $y = x$

- 13.1. What is the domain of f ? (1)
- 13.2. What is the range of f ? (1)
- 13.3. Write down the equation of
- 13.3.1. g (2)
- 13.3.2. h (state restriction) (3)
- 13.4. Sketch the graphs of g and h on the same set of axes, using the grid on the answer sheet provided. Clearly indicate the co-ordinates of the images of the points $(1; -1)$ and $(4; -2)$ on each graph. (6)
- 13.5. Describe the relationship between the graph of $y = f(x)$ and the graph of $y = \sqrt{x}$. (1)

QUESTION 14

The following sketch shows the graphs of:

- $f(x) = \log_a x$
- g - the reflection of f in the y -axis
- h - the reflection of f in the line $y = x$



- 14.1. Determine the value of a . (2)
- 14.2. Write down the co-ordinates of A and B. (2)
- 14.3. Write down the equation of g in the form $g(x) = \dots$ (1)
- 14.4. Determine the equation of h in the form $h(x) = \dots$ (2)

Functions: Solutions
Parabolas and Lines
Question 1

$$\begin{aligned}
 1.1. \quad & -3x^2 - 9x + 30 = 0 \\
 & x^2 + 3x - 10 = 0 \\
 & (x - 2)(x + 5) = 0 \\
 & x = 2 \quad \text{or} \quad x = -5
 \end{aligned}$$

$$\begin{aligned}
 AB &= 2 - (-5) \\
 &= \underline{7 \text{ units}},
 \end{aligned}$$

$$\begin{aligned}
 1.2. \quad & -12x + 12 = -3x^2 - 9x + 30 \\
 & 3x^2 - 3x - 18 = 0 \\
 & x^2 - x - 6 = 0 \\
 & (x - 3)(x + 2) = 0 \\
 & x = -2 \quad \text{or} \quad \underline{x = 3}, \\
 & \text{N.A}
 \end{aligned}$$

Sub $x = 3$ into $g(x)$

$$\begin{aligned}
 g(3) &= -12(3) + 12 \\
 &= \underline{-24},
 \end{aligned}$$

$$K(3, -24)$$

1.3.

$$f(x) - g(x) \leq 0$$
$$\downarrow$$
$$f(x) \leq g(x)$$

$$x \leq -3 \quad \text{OR} \quad x \geq 2$$

1.4.

$$CD = -3x^2 - 9x + 30 - (-12x + 12)$$
$$= -3x^2 - 9x + 30 + 12x - 12$$
$$= -3x^2 + 3x + 18$$

Method 1: Completing the square:

$$CD = -3(x^2 - x - 6)$$
$$= -3\left(x^2 - x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 - 6\right)$$
$$= -3\left[\left(x - \frac{1}{2}\right)^2 - \frac{25}{4}\right]$$
$$= -3\left(x - \frac{1}{2}\right)^2 + \frac{75}{4}$$

$\therefore$ Max length is $\frac{75}{4}$ units

Method 2: Derivative

$$CD' = -6x + 3 = 0$$
$$6x = 3$$
$$x = \frac{1}{2}$$

Sub $x = \frac{1}{2}$ into CD

$$CD = -3\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right) + 18$$
$$= \frac{75}{4} \text{ units}$$

Question 2

2.1. $x^2 - 4x + 3 = 0$
 $(x-3)(x-1) = 0$
 $x=3 \quad x=1$
 N.A

T(1, 0)

2.2. $f(x) = x^2 - 4x + 3$

Method 1: Completing the square

$$\begin{aligned} f(x) &= x^2 - 4x + (-2)^2 - (-2)^2 + 3 \\ &= (x-2)^2 - 1 \end{aligned}$$

$\therefore \underline{S(2, -1)}$

Method 2: Derivative.

$$f'(x) = 2x - 4 = 0$$

 $x=2$, sub into f

$$\begin{aligned} f(2) &= 2^2 - 4(2) + 3 \\ &= \underline{-1} \end{aligned}$$

$S(2, -1)$,

$$\begin{aligned}
 2.3. \quad PQ &= 3-x - (x^2 - 4x + 3) \\
 &= 3-x - x^2 + 4x - 3 \\
 &= -x^2 + 3x
 \end{aligned}$$

Method 1: Completing the square

$$\begin{aligned}
 PQ &= -(x^2 - 3x) \\
 &= -(x^2 - 3x + (-\frac{3}{2})^2 - (-\frac{3}{2})^2) \\
 &= -\left[(x - \frac{3}{2})^2 - \frac{9}{4}\right] \\
 &= \underline{-\left(x - \frac{3}{2}\right)^2 + \frac{9}{4}}
 \end{aligned}$$

$\therefore$ Max length PQ is $\frac{9}{4}$ units,

Method 2: Derivative

$$\begin{aligned}
 PQ' &= -2x + 3 = 0 \\
 x &= \frac{3}{2} \quad \text{sub into PQ}
 \end{aligned}$$

$$\begin{aligned}
 PQ &= -\left(\frac{3}{2}\right)^2 + 3\left(\frac{3}{2}\right) \\
 &= \underline{\frac{9}{4} \text{ units}}
 \end{aligned}$$

$$\begin{aligned}
 24. \quad p &= 2 \\
 q &= 1
 \end{aligned}$$

$$2.5 \quad 0 < x < 3$$

Question 3

3.1. $-3 \leq x \leq 2$

3.2. $f(x) = a(x+3)(x-2)$ sub $D(1; -8)$
 $-8 = a(1+3)(1-2)$
 $-8 = -4a$
 $\therefore \underline{a = 2}$

$$\begin{aligned}\therefore f(x) &= 2(x+3)(x-2) \\ &= 2(x^2 + x - 6) \\ &= 2x^2 + 2x - 12\end{aligned}$$

$$\therefore \underline{a = 2}, \quad \underline{b = 2}, \quad \underline{c = -12}$$

3.3. Method 1: Completing the square

$$\begin{aligned}f(x) &= 2(x^2 + x - 6) \\ &= 2\left(x^2 + x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 - 6\right) \\ &= 2\left[\left(x + \frac{1}{2}\right)^2 - \frac{25}{4}\right] \\ &= 2\left(x + \frac{1}{2}\right)^2 - \frac{25}{2}\end{aligned}$$

$$\therefore \text{Turning point } \left(-\frac{1}{2}; -\frac{25}{2}\right)$$

Method 2: Derivative

$$\begin{aligned}f'(x) &= 4x + 2 = 0 \\ x &= -\frac{1}{2}\end{aligned}$$

$$\begin{aligned}f\left(-\frac{1}{2}\right) &= 2\left(-\frac{1}{2}\right)^2 + 2\left(-\frac{1}{2}\right) - 12 \\ &= -\frac{25}{2}\end{aligned}$$

$$\text{Turning point: } \left(-\frac{1}{2}; -\frac{25}{2}\right)$$

(6)

3.4

$$f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{25}{2}$$

$$\begin{aligned} f(x-7) &= 2\left(x-7 + \frac{1}{2}\right)^2 - \frac{25}{2} \\ &= 2\left(x - \frac{13}{2}\right)^2 - \frac{25}{2} \end{aligned}$$

$$\begin{aligned} h(x) &= f(x-7) + 2 \\ &= 2\left(x - \frac{13}{2}\right)^2 - \frac{25}{2} + 2 \\ &= 2\left(x - \frac{13}{2}\right)^2 - \frac{21}{2} \end{aligned}$$

$$\therefore \underline{x = \frac{13}{2}},$$

3.5.

$$f'(x) = 4x + 2$$

$$f'(1) = 4(1) + 2$$

$$= \underline{6},$$

$$\therefore \underline{m = 6},$$

HyperbolasQuestion 4

$$4.1 \quad x = -2$$

$$y = -1$$

4.2.1.

$$\begin{aligned} y &= \frac{6}{0+2} - 1 \\ &= \underline{2}, \end{aligned}$$

4.2.2

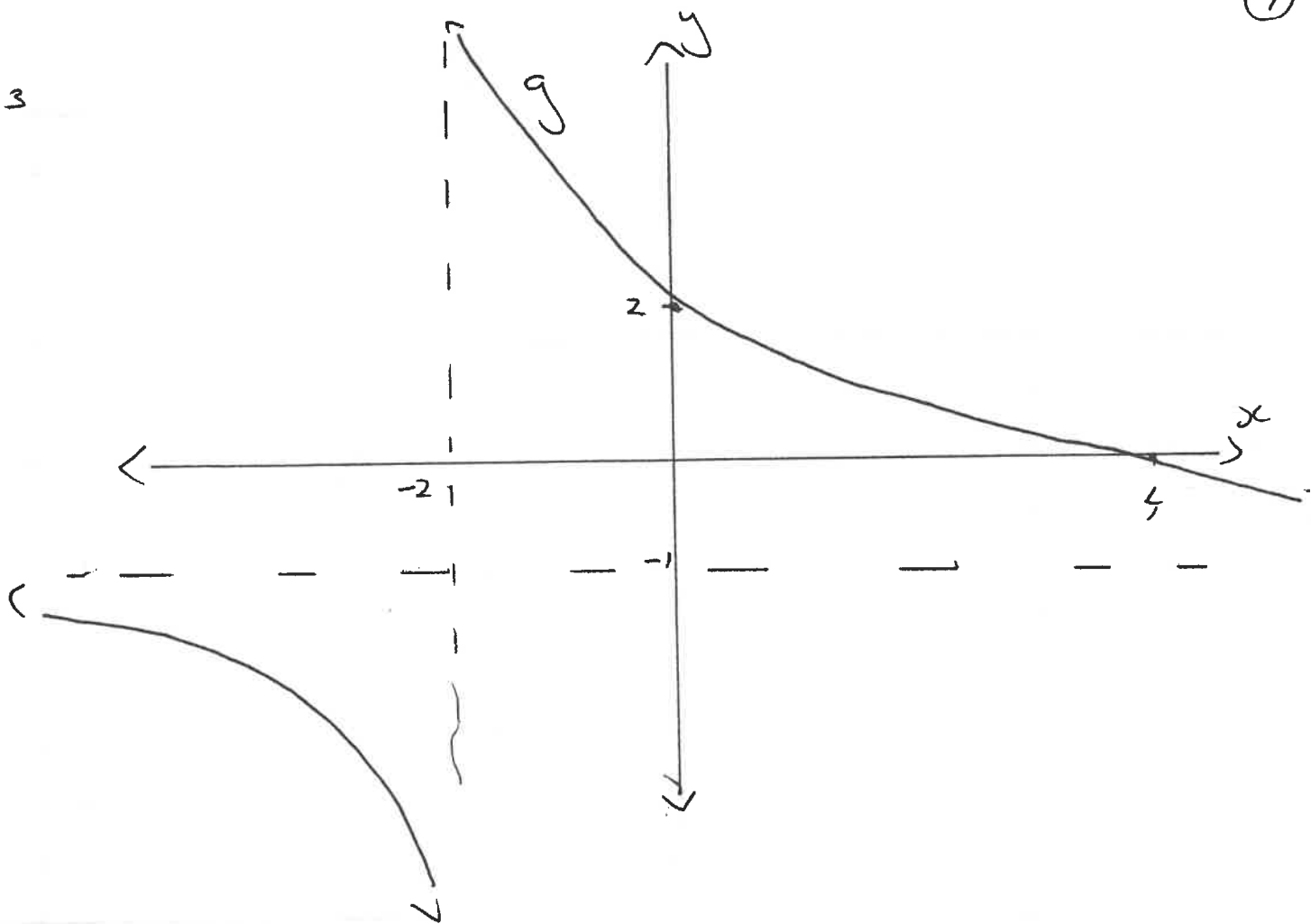
$$\frac{6}{x+2} - 1 = 0$$

$$\frac{6}{x+2} = 1$$

$$6 = x+2$$

$$\underline{4 = x},$$

4.3



4.4.

$$\begin{aligned}
 y &= -x + c \\
 -1 &= -(-2) + c \\
 -3 &= c
 \end{aligned}$$

sub $(-2, -1)$

$$\therefore \underline{y = -x - 3},$$

4.5.

$$x \in (-2; \infty)$$

$$\text{or } -2 < x < \infty$$

Question 5

$$5.1. \quad y \in \mathbb{R}; \quad y \neq -2$$

$$5.2. \quad y = \frac{a}{x-1} - 2$$

$$-5 = \frac{a}{0-1} - 2$$

$$-3 = -a$$

$$\underline{a = 3}$$

$$5.3. \quad h(x) = \frac{3}{(x-3)-1} - 2 + 7$$

$$= \frac{3}{x-4} + 5$$

$\therefore$ Coordinates: (4; 5)

Question 6

$$6.1. \quad y = 1$$

$$6.2. \quad x \in \mathbb{R}; \quad x \neq -2$$

$$6.3. \quad y = -x + c \quad \text{sub } (-2; 1)$$

$$1 = -(-2) + c$$

$$1 = 2 + c$$

$$-1 = c$$

$$\therefore y = -x - 1$$

$$\underline{h(x) = -x - 1}$$

6.4.1. f is reflected in the x -axis and moved 3 units up.

6.4.2. $y \in \mathbb{R}; y \neq 2$

6.5 $y = mx$ sub $(-2; 1)$
 $1 = m(-2)$
 $m = -\frac{1}{2}$

$y = \frac{k}{x+2} + 1$ sub $(0; 0)$

$0 = \frac{k}{0+2} + 1$

$-1 = \frac{1}{2}k$

$-2 = k$

Question 7:

7.1. $x \in \mathbb{R}; x \neq -2$

7.2.1. $\frac{x-3}{x+2} = 0$

$x-3 = 0$

$x = 3$

$\therefore \underline{(3; 0)},$

7.2.2. $y = \frac{0-3}{0+2}$

$y = -\frac{3}{2}$

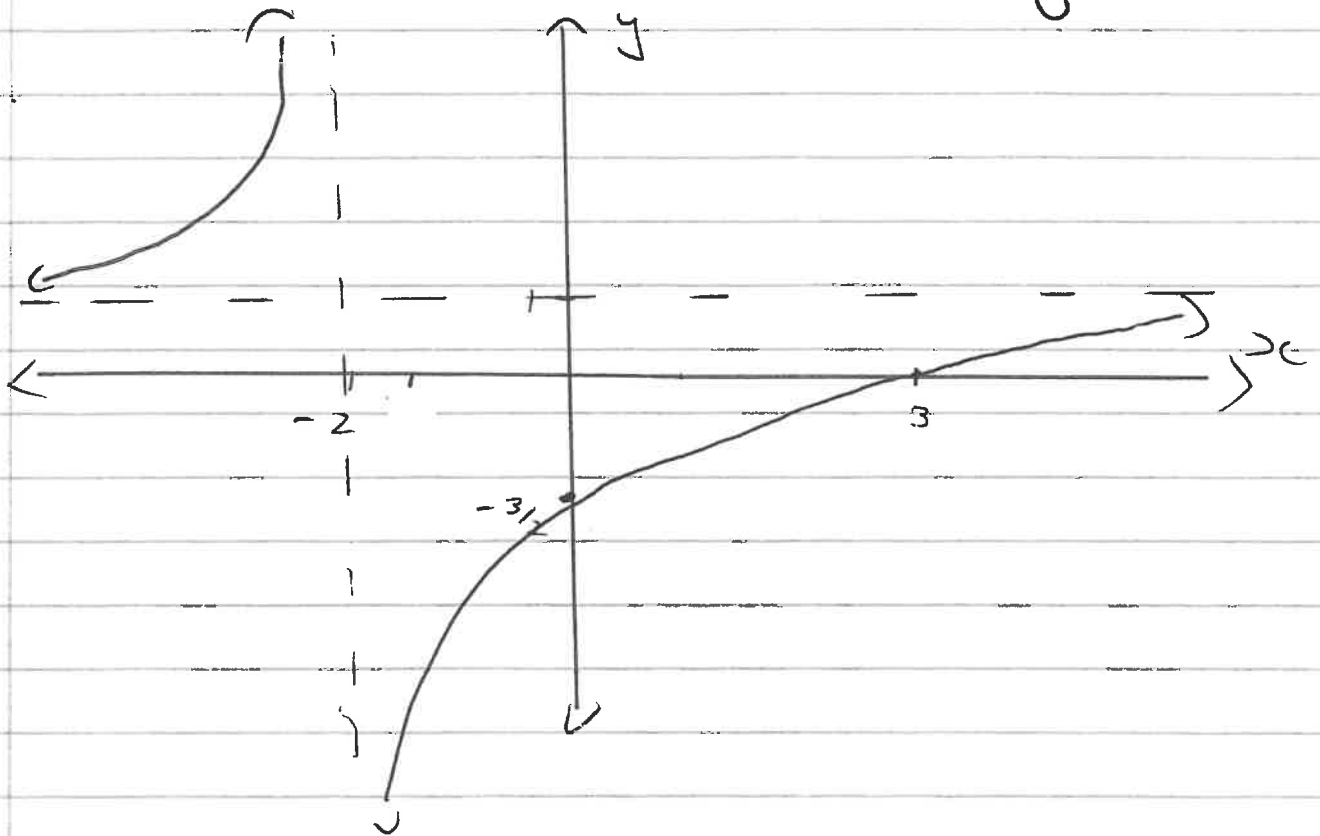
$\therefore \underline{(0; -\frac{3}{2})},$

7.3.

$$\begin{aligned}
 f(x) &= \frac{x-3}{x+2} \\
 &= \frac{x+2-5}{x+2} \\
 &= \frac{x+2}{x+2} - \frac{5}{x+2} \\
 &= 1 - \frac{5}{x+2} \\
 &= \frac{-5}{x+2} + 1
 \end{aligned}$$

$\therefore y = 1$

7.4.



7.5.

$-2 < x \leq 3$

Exponential Functions

Question 8

8.1.

$$q = 1$$

$$\therefore y = -(3)^{x+p} + 1 \quad \text{sub } (1, 0)$$

$$0 = -(3)^{1+p} + 1$$

$$-1 = -(3)^{1+p}$$

$$1 = 3^{1+p}$$

$$3^0 = 3^{1+p}$$

$$0 = 1+p$$

$$p = -1$$

$$\therefore \underline{f(x) = -(3)^{x-1} + 1}$$

8.2.

$$\begin{aligned} k(x) &= -(3)^{x-1-2} + 1 - 3 \\ &= \underline{-(3)^{x-3} - 2} \end{aligned}$$

8.3

$$\begin{aligned} l(x) &= -f(x) \\ &= -[-(3)^{x-1} + 1] \\ &= \underline{3^{x-1} - 1} \end{aligned}$$

Question 9

9.1.

$$y = -3$$

9.2.

$$x \in \mathbb{R} ; x \neq 1$$

9.3.

$$b = -3$$

$$\text{Find } p: \quad \frac{3}{x-1} - 3 = 0$$

$$\therefore \frac{3}{x-1} = 3$$

$$\frac{1}{x-1} = 1$$

$$1 = x-1$$

$$\underline{x = 2},$$

$$P(2; 0)$$

Sub into

$$y = a^x - 3$$

$$0 = a^2 - 3$$

$$3 = a^2$$

$$\therefore \underline{a = 3^{\frac{1}{2}}}$$

$$9.4. \quad y \in (-3; \infty)$$

$$9.5.1. \quad -1 < x \leq 2$$

$$9.5.2. \quad -\infty < x < 1$$

Inverses and Symmetries / Reflections

Question 10

10.1. Function becomes one-to-one.
Inverse is a function.

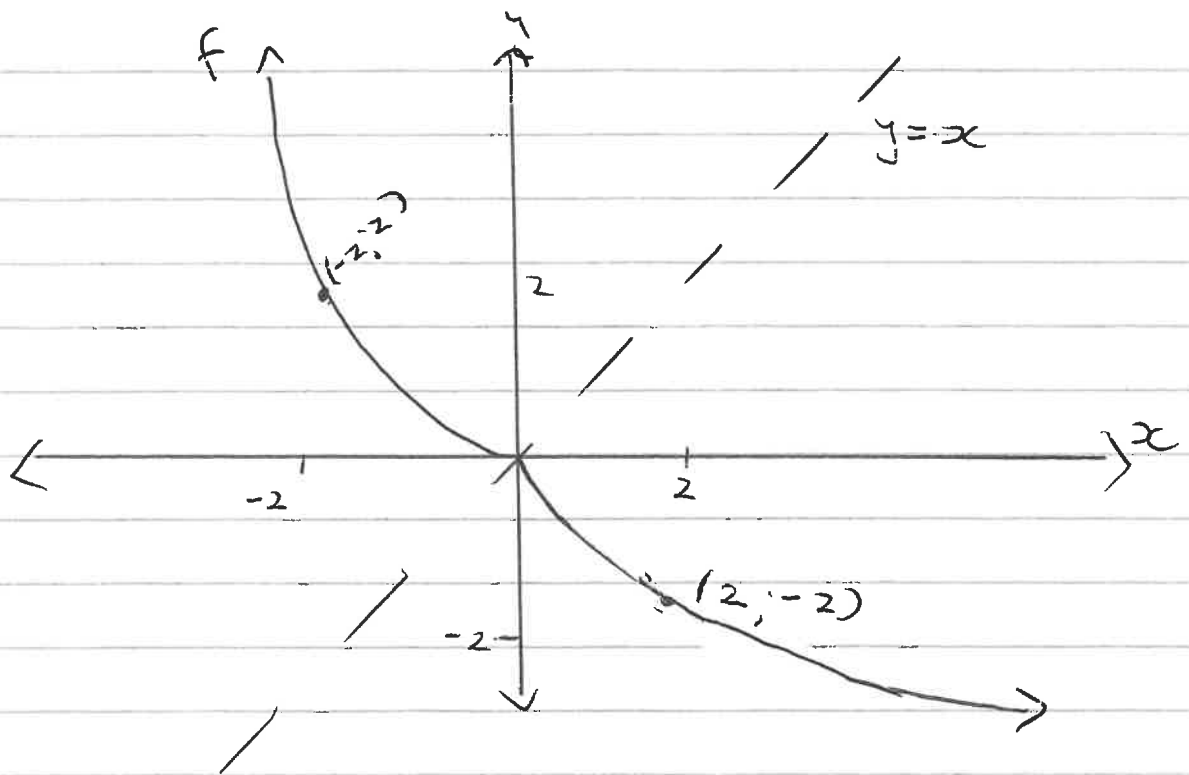
$$10.2. \quad y = \frac{1}{2}x^2 \quad ; \quad x = y$$

$$x = \frac{1}{2}y$$

$$2x = y^2$$

$$y = -\sqrt{2x} \quad ; \quad x \geq 0$$

10.3



10.4.

$$\begin{aligned}
 g(x) &= -f(x) \\
 &= -\left(\frac{1}{2}x^2\right); \quad x \leq 0 \\
 &= \underline{-\frac{1}{2}x^2; x \leq 0}
 \end{aligned}$$

Question 11

11.1

$$\begin{aligned}
 f(-2) &= \left(\frac{1}{4}\right)^{-2} \\
 &= \underline{16},
 \end{aligned}$$

11.2

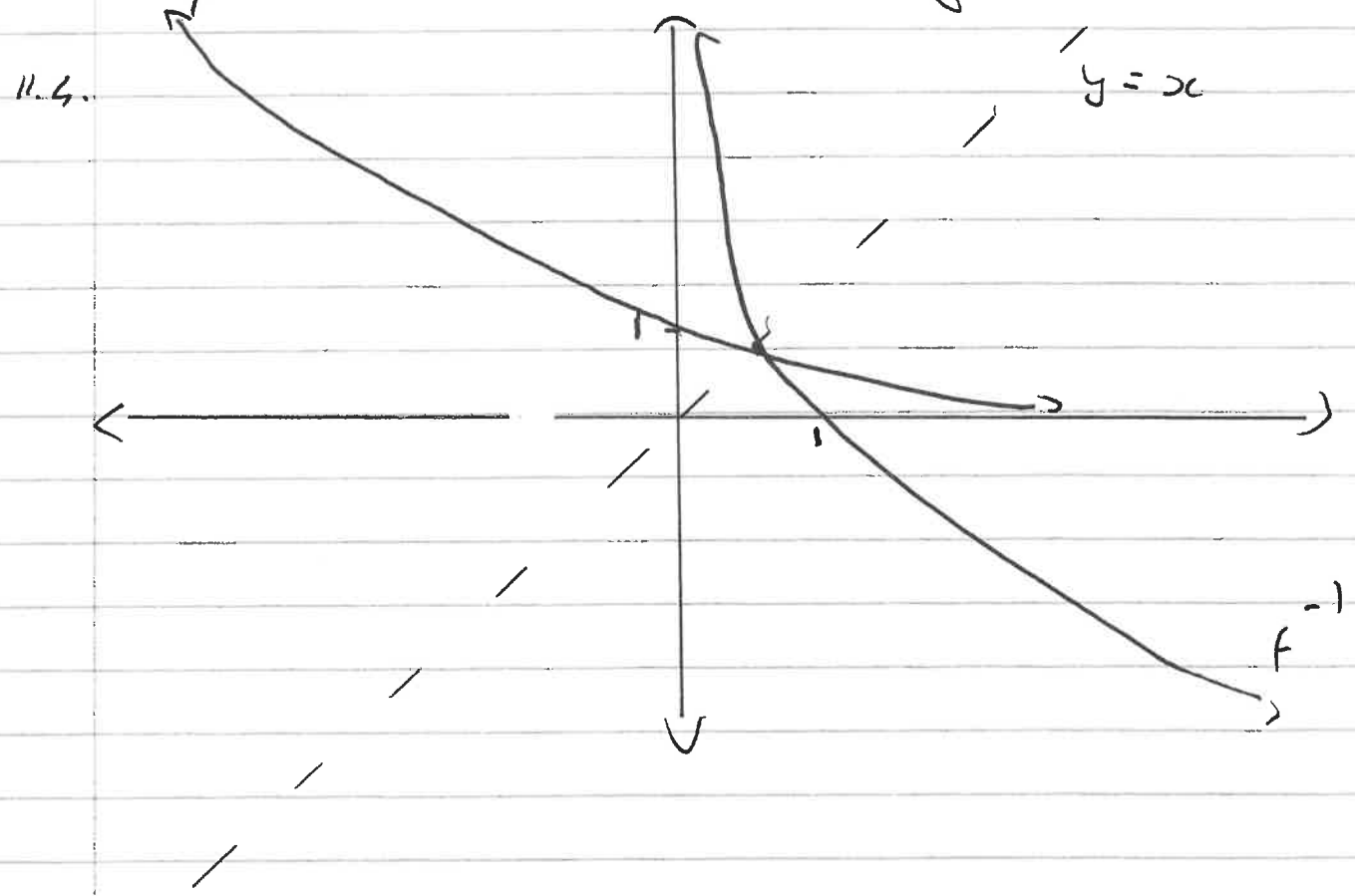
$$y = \left(\frac{1}{4}\right)^x; \quad x = y$$

$$x = \left(\frac{1}{4}\right)^y$$

$$y = \log_{\frac{1}{4}} x$$

$$\underline{f^{-1}(x) = \log_{\frac{1}{4}} x}$$

11.3 Reflect f about the line $y=x$



11.5 $x \in (0; \infty)$ OR $x > 0$

11.6 $0 < x \leq 16$

11.7.1.

$$q = \log_{\frac{1}{2}} \frac{1}{2}$$
$$\frac{1}{4} q = \frac{1}{2}$$
$$\left(\frac{1}{2}\right)^{2q} = \frac{1}{2}$$
$$2q = 1$$
$$q = \frac{1}{2}$$

11.7.2.

$$x = \log_{\frac{1}{2}} x$$
$$\therefore \left(\frac{1}{2}, \frac{1}{2}\right)$$

Question 12:

12.1.

$$a^{x-2} = 9 a^x$$

$$\frac{a^{x-2}}{a^x} = 9$$

$$a^{-2} = 9$$

$$\underline{a = \frac{1}{3}},$$

12.2.1.

$$f(x) = \frac{1}{3}^x$$

$$y = \frac{1}{3}^x$$

$$; \quad y = x$$

$$x = \frac{1}{3}^y$$

$$y = \log_{\frac{1}{3}} x$$

$$\underline{f^{-1}(x) = \log_{\frac{1}{3}} x}$$

→

12.2.2.

$$y = x$$

12.2.3.

$$\underline{t = 3},$$

Question 13:

13.1.

$$x \in [0; \infty)$$

OR

$$x \geq 0$$

13.2.

$$y \in (-\infty; 0]$$

OR

$$y \leq 0$$

13.3.1.

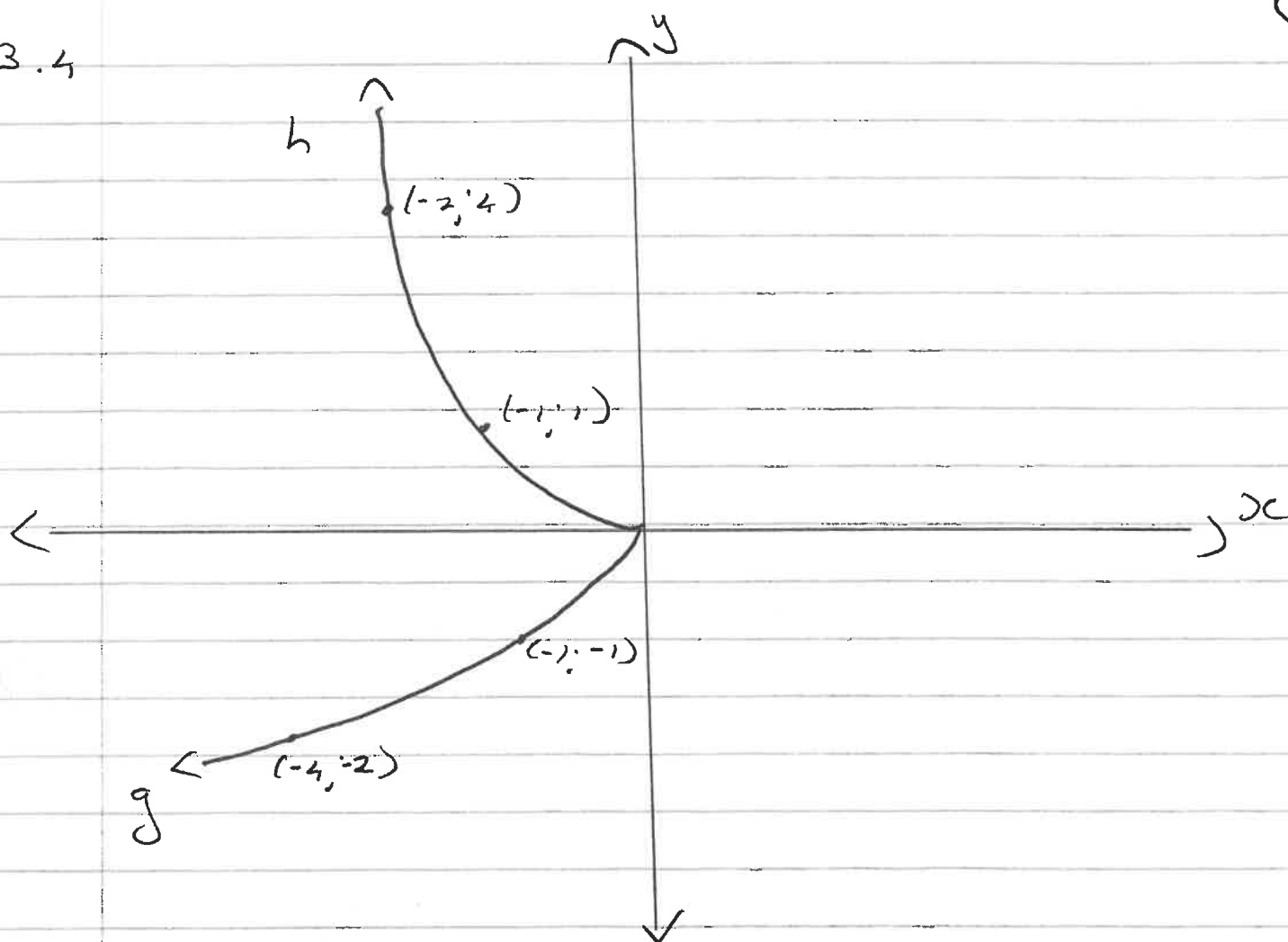
$$g(x) = -\sqrt{-x}$$

$$x \leq 0$$

13.3.2.

$$h(x) = x^2 ; \quad x \leq 0$$

13.4



(16)

13.5 f Reflected in the x-axis is $y = \sqrt{x}$

Question 14:

14.1. $y = \log_a x$ sub $(9, 2)$

$$2 = \log_a 9$$

$$a^2 = 9$$

$$\underline{a = 3}$$

14.2. A $(1, 0)$
B $(0, 1)$

14.3. $g(x) = \log_3(-x)$; $x < 0$

14.4. $h(x) = 3^x$